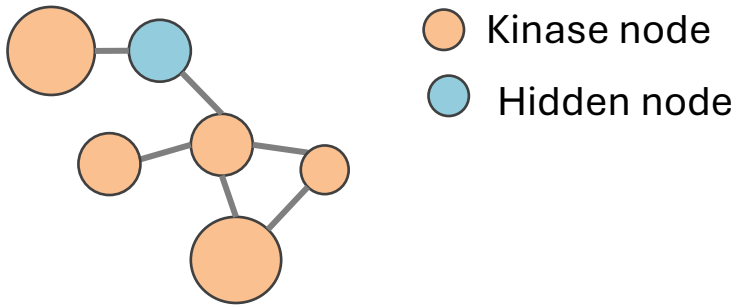
Kinase networks and pathways

Interpretation guide

Quick start

How are networks generated?

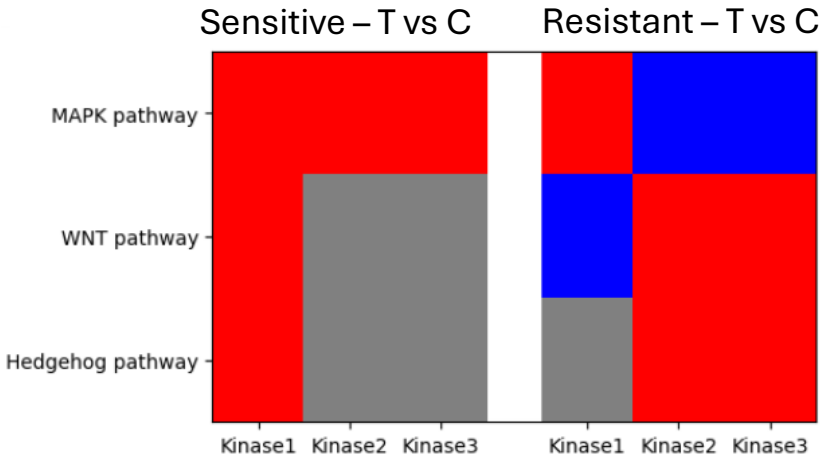
Networks are generated from high scoring* kinases. Networks connect key kinases through high-confidence links and may include high confidence connector (hidden) nodes which are not in the data.



Files:

Network.html: an interactive network figure which shows the Kinase Statistic (LogFC) and the pathways. Nodes and pathways are searchable. Networks help to understand kinases relationships.

In the network, each cluster of nodes is annotated with pathways by EnrichR, using the Reactome db.



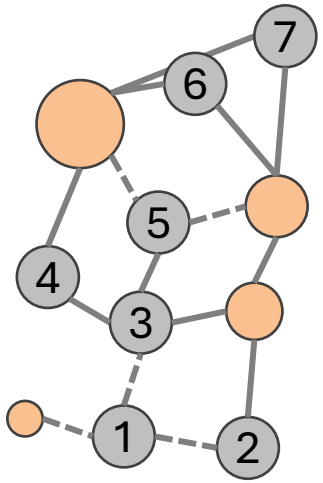
Files:

Kinase_pathway_heatmaps folder: start exploring pathways with these figures to assess the change in the pathways across comparisons.

***Params.csv**: shows how the data was pre-filtered. spec_cutoff is the threshold for specificity score. perc_cutoffs are the threshold(s) for percentile, which is usually 0.01 (shown at the end of the filename as _p01) so everything is used above the specificity cutoff.

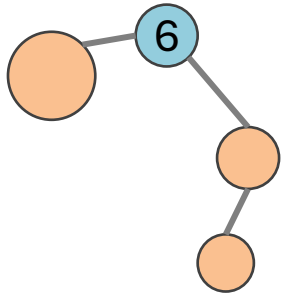
Network generation enhances UKA results

INPUTS:
- STRING as reference network;
  - selected nodes from UKA



PCSF algorithm

RESULT:
"OPTIMAL" SUBNETWORK



- **Key kinases & high-confidence links**
- Less FP (false positive – weakly connected)
- Less FN (false negative, hidden nodes)

Nodes

Input nodes (size ~ LFC)

Hidden nodes

High evidence edge

Low evidence edge

Generating networks enhances robustness of UKA results:

- pruning weakly connected or low-fold-change nodes (probably FP)
- and recovering missing connections through hidden nodes (probably FN).

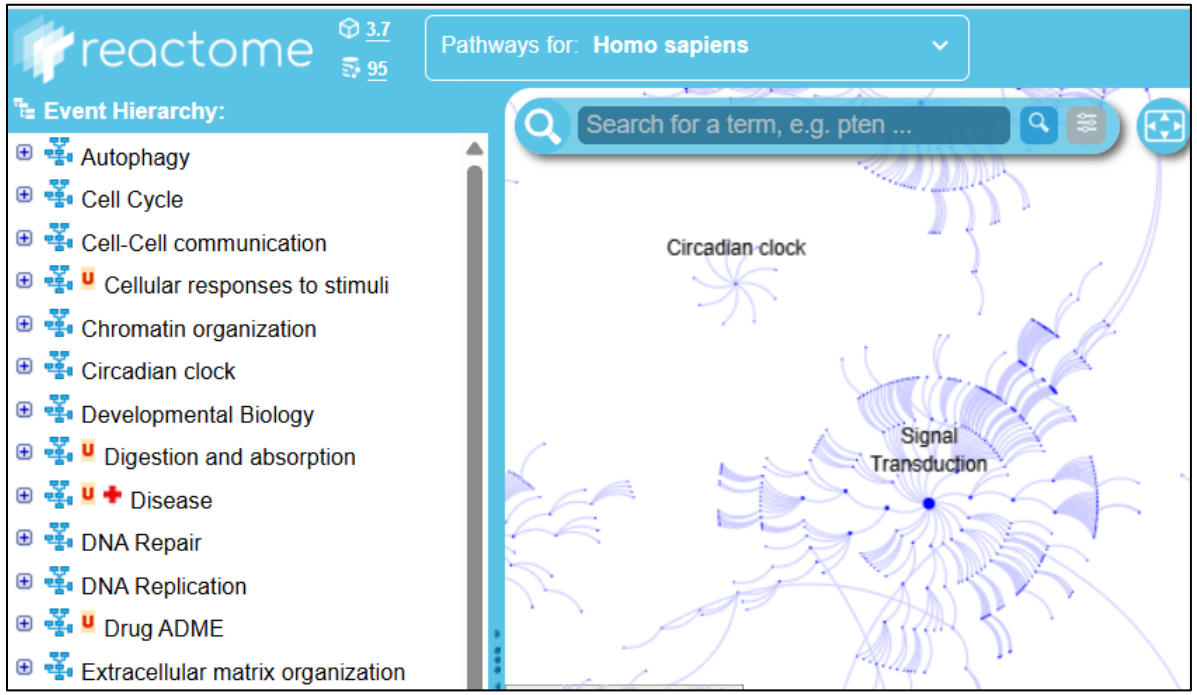
The Prize-Collecting Steiner Forest (PCSF) algorithm, given the reference network and input nodes, constructs a subnetwork through optimizing to:
- include more input nodes with high prize (prize = absolute LFC)
- Decrease edge costs (low cost ~ high evidence).

Based on (Arici & Tuncbag, 2021)

How to select most relevant pathways?

Use the Pathways_<T vs C>.csv:

- Every pathway belongs to a Reactome pathway group (e.g. AKT signaling is under Signal Transduction).
- These columns can help to identify relevant pathways:
  - *N_hits, pathway_size, overlap columns* show how many pathway members are in the data. (e.g. n_hits = 3, pathway_size = 10 means 3 kinases from the results are found in a pathway of 10 members).
  - *Hierarchy_n column*: shows the hierarchy level (e.g. Signal Transduction is level 1). Top levels are generic, bottom levels are specific. Middle levels might be the most interesting.



The **Pathways_<T vs C>.csv** file lists the selected pathways for each gene cluster (column Genes). However, additional pathways may also be annotated to the same gene cluster. In this file, the pathway chosen is the highest in the hierarchy (i.e., the most general). If a pathway is irrelevant, use **Pathways_<T vs C>_all.csv** to check whether there is another, more relevant pathway for the same gene cluster.

How to select most relevant pathways?

Which are the most relevant pathways?
There is no simple metric. Pathways usually share members, and a pathway that is relevant in one model may be irrelevant in another.
In addition, statistical tests do not consider how specific members are to a pathway. As a result, a pathway may receive a strong p-value even if only non-specific members are detected.

Therefore, we recommend assessing pathway relevance using biological domain knowledge rather than relying on p-values or combined scores.

Other network files

- **Nodes.csv:** list of all kinase and hidden nodes in the network
- **Nodes_clusters:** Network clusters.
- **Missing nodes.csv:** kinases that passed the thresholds but are not in the network because they are weakly connected / low fold change. Might be false positives.
- **Nodes with pathways.csv:** nodes to pathways mapping
- Folders with the files of the network html (don’t delete it)

PCSF in details

The PCSF (Prize-Collecting Steiner Forest algorithm) solves a network optimization problem. Given a reference network, where the nodes have prizes and the edges have costs, it aims to identify a subnetwork with a given node input. Subnetwork construction is an optimization to minimize the total edge costs and the total node prizes left out:

$$Minimize\left(\sum edge\ cost + b\sum prize\ of\ nodes\ left\ out\right)$$

Where b is used to tune the prizes of nodes relative to edge costs. Higher b leads to more input nodes incorporated in the result subnetwork.

We use the PCSF R package with the STRING network (v12.0) containing all edges as the reference interactome. Node prizes are assigned based on the UKA and Target Fold Changes.

Evidence channels of STRING

Evidence Channel	Type	Reliability	Key Strength
Experimental	Direct	🥇 Highest	Proof of direct physical or genetic interaction.
Curated Databases	Direct	🥇 Very High	Expert-validated functional association in pathways/complexes.
Gene Fusion	Genomic	🥈 High	Strong evolutionary evidence for functional interaction.
Gene Neighborhood	Genomic	🥈 High (Prok) / Low (Euk)	Evidence for operons and co-regulation (mainly in bacteria).
Gene Co-occurrence	Genomic	🥈 Medium-High	Evidence for participation in the same pathway/system.
Co-expression	Functional	🥈 Medium	Strong evidence for co-regulation and functional association.
Text mining	Literature	🥉 Lower	High-level associative